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QUESTION 5 • UNIT 2 – TRAINING METHODS

Explain how a Neural Network can be trained with Back propagation and Forward propagation methods. Compare both methods.

📝 5 Marks

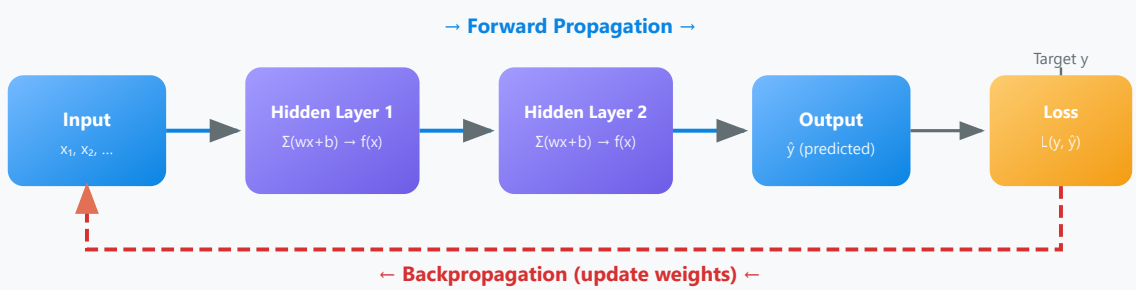
Forward Propagation & Backpropagation

Training a neural network — step-by-step with diagrams and comparison

Overview

Training a neural network involves two key processes: **forward propagation** and **backpropagation**. These methods work together to optimize the network's weights and biases to **minimize the error** between predicted and actual outputs.

Figure 1 — The Complete Training Cycle



Forward propagation flows left → right (input → output). Backpropagation flows right → left (output → input), updating weights to reduce error.

1. Forward Propagation

Forward propagation is the process of passing input data through the layers of a neural network to **compute the output**. Data flows in one direction — from input to output.

Steps in Forward Propagation

1 Input Layer

Input data is fed into the network. Each input is multiplied by its corresponding weight and added to a bias term.

2 Hidden Layers

The weighted sum of inputs is passed through an **activation function** to introduce non-linearity. The output of one layer becomes the input for the next layer.

$$a^{[l]} = f(W^{[l]} \cdot a^{[l-1]} + b^{[l]})$$

3 Output Layer

The final layer produces the output $\hat{y}$, which is compared to the actual target value using a **loss function**.

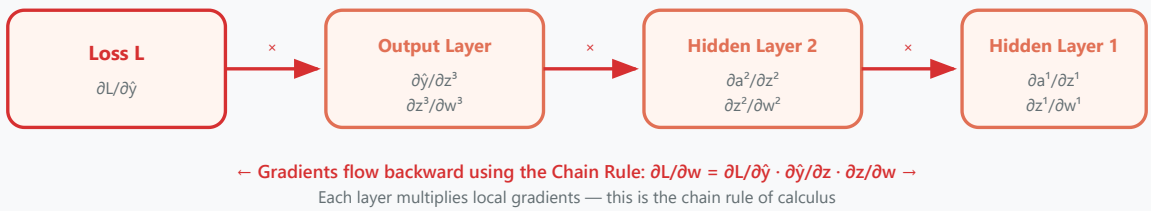
Key Points:

- ✓ Forward propagation calculates the predicted output.
- ✓ It does **not** involve any weight updates.
- ✓ It is a feed-forward process — data flows in one direction (input → output).

2. Backpropagation

Backpropagation is the process of calculating the **gradient of the loss function** with respect to the weights and biases in the network, and then updating these parameters to minimize the loss.

Figure 2 — Chain Rule in Backpropagation



Backpropagation uses the chain rule to compute gradients layer by layer, from the loss back to the first hidden layer.

Steps in Backpropagation

1 Calculate Error

The error is computed using a loss function (e.g., MSE, Cross-Entropy) by comparing the predicted output $\hat{y}$ to the actual target value y.

2 Compute Gradients

The gradient of the loss function with respect to each weight and bias is calculated using the **chain rule of calculus**. Gradients indicate the direction and magnitude of change needed to minimize the loss.

3 Update Weights and Biases

Weights and biases are updated using an optimization algorithm (e.g., Gradient Descent):

$$w^{(new)} = w^{(old)} - \alpha \cdot \frac{\partial L}{\partial w}$$

Where α is the learning rate, L is the loss function, and w is the weight.

4 Repeat

The process is repeated for **multiple epochs** until the loss is minimized and the model converges.

Key Points:

- Backpropagation adjusts the weights and biases to minimize the error.
- It is a **feedback process** — data (gradients) flows backward from the output to the input.

Comparison: Forward Propagation vs. Backpropagation

Aspect	Forward Propagation	Backpropagation
Direction	Data flows **forward** (input → output)	Error flows **backward** (output → input)
Purpose	Computes the predicted output $\hat{y}$	Updates weights & biases to minimize error
Math Process	Matrix multiplication + activation functions	Gradient calculation using the chain rule
Role in Training	Provides output for comparison with target	Optimizes network parameters based on error
Dependency	Independent of backpropagation	Requires output from forward propagation
Output	Predicted values $\hat{y}$	Updated weights and biases

Aspect	Forward Propagation	Backpropagation
Weight Update	No weight updates occur	Weights are updated every iteration

Summary

- ✓ **Forward Propagation** is the process of calculating the output of the network based on the input and current weights.
- ✓ **Backpropagation** is the process of adjusting the weights and biases to minimize the error between predicted and actual outputs.
- ✓ Both methods are **essential** for training a neural network — forward propagation computes the output, and backpropagation ensures the model learns from its mistakes.